

Motion Planning Task



Name: Natalie Sameh Saad

Code: 2500617

Report: Questions answers

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Question 1:

- Industrial automation: It performs tasks as material handling, assembly and inspection to increase productivity and reduce costs.
- Logistics and Warehousing: It is used in them to pick orders, inventory management and transportation of good with high speed and efficiency of operations.
- Agriculture: it is used in crop monitoring, spraying pesticides and harvesting to improve yields and decrease costs.
- Healthcare: it is used in patient monitoring, medication delivery to reduce risk of infections and improve patient care.
- Search & Rescue: it is used in search and rescue operations in disaster-stricken areas to rescue victims for a safer and more efficient approach.
- Environmental Monitoring: it is used in air and water quality monitoring, weather forecasting and pollution control for more accurate data.
- Security & Surveillance: it is used in perimeter patrol, crowd monitoring and intrusion detection for safety and security.
- Entertainment: it is used in theme parks and exhibitions, interactive displays, games to enhance visitor experience.
- Space Exploration: it is use in exploring remote planets and moons , gathering data and these tasks are too risky to be performed by human.
- Education & Research: it is used in teaching programming and robotics for experiments and gathering data for interactive approach.

Question 2:

Algorithm Category	Approach	Strengths	Weakness
Exact algorithms	Finding the exact solution of path planning problem by exploring all possible paths.	Finding optimal solution even in complex environments.	can be computationally expensive and time-consuming in large-scale environment.
Heuristic algorithms	Using a set of heuristics or rules to guide search for an optimal path and reducing the search space.	Efficient and suitable for large-scale environments finding suboptimal solution quickly.	May not find the optimal solution and get stuck in local minima.
Probabilistic algorithms	using a probabilistic model to generate a set of feasible, collision-free paths, considering the environment and the robot's dynamics.	handling uncertainty in the environment, suitable for dynamic and unpredictable environments.	Can be computationally expensive and may require a large number of samples to find a good solution.
Hybrid algorithms	Combining 2 or more of the above approaches to improve the performance and accuracy of the path planning process.	Combine the strengths of different techniques and can provide a better solution than using a single technique.	Can be complex to be applied and may require extensive tuning parameters.

Question 3:

Exact algo. Steps: 1. Represent the environment using a graph-based representation as nodes represent the points and edges represent the feasible transitions between the points. 2. Defining the starting point and the destination of the robot. 3. Compute the cost function to measure distances or time needed to travel between 2 points. 4. Choosing the search algorithm to get the shortest distance. 5. Execute the algorithm with calculating cost of travelling and choosing the best path. 6. Implementing the optimal path avoiding collisions. 7. Continuous update the path to ensure it remains optimal away from obstacles. It is suitable for simple and defined environment where optimal solution is needed neglecting speed of getting it in industry.

Heuristic algo. Steps: 1. Define the problem (start and end point, obstacles and any constraints). 2. Define heuristic function by estimating cost of reaching goal from any point and is admissible. 3. Generate a search tree of all possible paths between start and ending points. 4. Choose a search strategy. 5. Apply heuristics to direct the search. 6. Optimize the solution by refining search tree or adjusting the function. 7. Implement the solution using control algorithm to navigate robot. It is suitable for large-scale environment where speed is more essential than optimality as warehouse automation.

Probabilistic algo. Steps: 1. define the environment (obstacles and boundaries). 2. Define start and goal states. 3. Generate a roadmap by random sampling the space and check for collisions. 4. Search for a feasible path by searching algorithms to evaluate potential of paths and get the shortest one. 5. navigate the robot using control algo. as PID. 6. Update the roadmap by adding or removing nodes according to changes in environment. 7. repeat previous steps to update roadmap and find feasible paths. It is suitable for dynamic and unpredictable environment as map data is unavailable for navigation.

Hybrid algo. Steps: 1. Identify the requirements of applications to guide the selection and design of the algorithm. 2. select 2 or more suitable algorithms as one for high accuracy and the other for faster planning. 3. identify the strength and weakness of each algorithm. 4. design the hybrid algorithm as how they are

combined using different techniques. 5. test the hybrid algorithm by simulations , real world experiments. 6 refine and optimize this algo. according to result to improve performance. It is used in complex, real-world problems with many obstacles including dynamic ones as single approach will fail to get results.

Question 4:

Dijkstra algorithm (Exact Example): It evaluates the cumulative cost of available paths, always prioritizing the route with the lowest accumulated cost. It moves step-by-step, adding up these costs until it reaches the final goal, finalizing the route with the absolute lowest total cost. it find the best optimal shortest path but consumes a lot of memory as it explore unnecessary paths and become slow. It can be used in static and closed environment that cares more about finding optimal paths not its speed as robotic arms used in industry.

A* algorithm(Heuristic Example): It directed towards the target by using a cost function that calculates the distance traveled and a heuristic estimate of the remaining distance to the goal. It finds the optimal shortest path very fast but if the heuristic function isn't accurate it will get stuck in local minima. It is used in large environments that need efficiency and speed as satellite navigation (as GPS) uses this algorithm for real-time traffic delays and speed limits.

RRT & RRT* (Probabilistic Example): RRT builds a search tree by generating random samples in the free space and growing branches from the nearest existing node toward them, rapidly exploring unknown areas. RRT* enhances this by adding a rewiring step and evaluates neighboring nodes in a radius and reconnects them if the new branch offers a lower cumulative cost. It works well in complex and 3D environments as it doesn't make any grid lines as RRT is very fast to find initial path and RRT* searching for the optimal shortest path over time and that is known as asymptotic optimality. RRT produces irregular paths that requires smoothing so RRT* solves this but consumes more computational power and time to rewire the branches. When maps are unavailable with complex obstacles in highly dynamic or high-dimension environment we can the method be used as Drones navigating through dense forests in 6D.

A*+RRT (hybrid=heuristic+ probabilistic): A* starts as global planner mapping the fastest path depending on grid lines which may make zigzagged lines then RRT acts as local planner in continuous spaces generates random nodes to break free from grid's limits and dodge any sudden obstacle as A* can't see. It's good in combining speed of finding path by A* with RRT ability to handle complex environments with obstacles and reduce time planning. But it's complex to be applied as it requires continuous data translation between grid based method to continuous space method. It is good for unpredictable environment that general map is known but any sudden obstacle can appear as search and rescue robot in a damaged building it follows the fast A* path but switches to RRT when sudden obstacle or rubble appears.

Question 5:

- 1- **Scalability:** As obstacles, dimensions and environmental complexity increase, algorithm computation becomes prohibitive, severely limiting real-time performance. It can be solved by apply parallel computing techniques such as GPU acceleration, distributed computing or multi-core processors to handle larger calculations rapidly.
- 2- **Uncertainty:** Variable sensor data due to noise and unpredictable robot motion often result in unsafe or suboptimal paths. It can be solved by integrating probabilistic models like Monte Carlo methods or Bayesian inference to mathematically account for sensor and environmental variability.
- 3- **Multiple Objectives:** Real-world navigation requires balancing speed, energy efficiency and safety simultaneously, as most algorithms optimize only one parameter as the shortest or the safest or maybe energy-efficient. It can be solved by utilizing multi-objective optimization models as Pareto-based approaches that generate a spectrum of optimal solutions weighing different constraints.
- 4- **Adaptability:** Robots frequently encounter entirely unexpected obstacles not represented on their initial maps, and traditional algorithms struggle to adjust immediately in real-time. It can be solved

employing deep learning and machine learning strategies that allow the system to adapt and learn dynamically from new obstacles and ones appears suddenly.

Question 6:

I found this paper very useful and informative. It broadened my understanding of how the algorithms are classified as I liked this part since I was doing the solomission and now I have understood more and more .However I noticed inaccuracies and contradictions in the paper in classification of specific algorithm as wrong examples in exact and probabilistic algorithms so I didn't rely only on the paper ,instead I made a research to identify these examples. Overall this paper was a deeply rewarding experience that increased my knowledge about path planning.